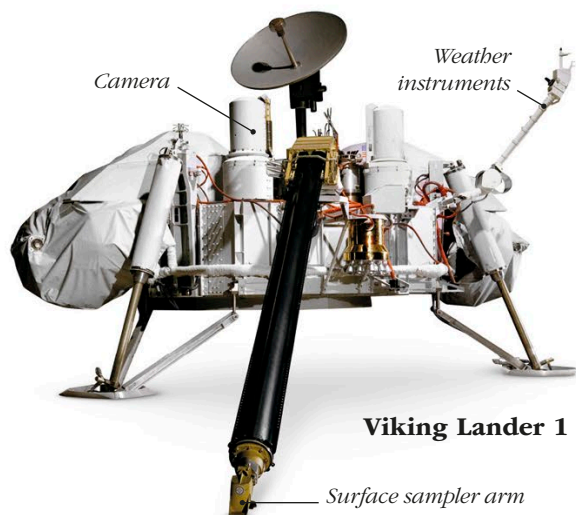


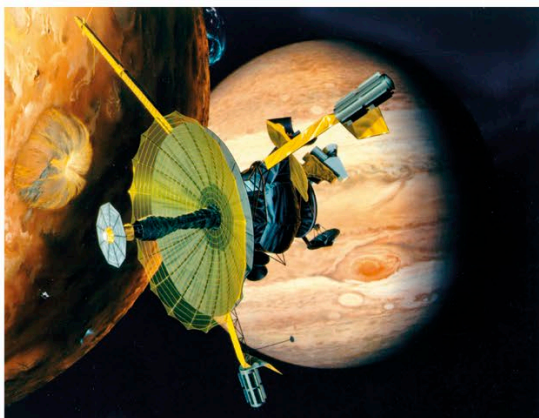


Viking Lander 1

LANDERS ON MARS

Mars is easier to reach than most other planets, but landing in its thin atmosphere is difficult. In 1976, NASA's pair of Viking orbiters put two landers on the surface, using parachutes and precisely fired retrorockets to control their descent. While the orbiters mapped Mars from space, the landers relayed the first images and data from the ground.

Illustration of Galileo and Jupiter



PROBING THE GIANTS

Following its first brief flybys of the giant outer planets, NASA has sent two orbiters to investigate Jupiter, and one to Saturn. Arriving at Jupiter in 1995, Galileo released a smaller probe that parachuted into the giant planet's atmosphere. The Cassini mission carried a European-built lander that touched down on Saturn's giant moon Titan in 2005.

► CURIOSITY ON MARS

NASA's car-sized, solar-powered Curiosity rover touched down on Mars in August 2012 and has since driven more than 11.2 miles (18km).

MARTIAN ROVERS

Since 1997, NASA has sent a series of increasingly ambitious robot rovers to explore Mars, investigating its atmosphere and geology and looking for signs of ancient life and water. Driving conditions on the rocky, dusty surface can be treacherous, and the limited speed of radio signals makes it difficult to direct the vehicles from mission control on Earth. So, recent rovers use artificial intelligence systems to make basic decisions—such as avoiding obstacles and recognizing which types of rock to study—without human intervention.

